MIL-C-20533
7 December 1951

MILITARY SPECIFICATION

CASES, CARTRIDGE, STEEL, PERFORATED, FOR ARTILLERY AMMUNITION

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Army 50-46-5
30 September 1947

This specification consists of this cover sheet, Specification 50-46-5, dated 30 September 1947, and Amendment 1 dated 19 April 1948, attached hereto, without modification.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer .

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No. 50-46-5

30 SEPTEMBER 1947

U. S. ARMY
SPECIFICATION

CASES, CARTRIDGE, STEEL, PERFORATED, FOR ARTILLERY AMMUNITION

A. APPLICABLE SPECIFICATIONS

A–1. The Ordnance Department drawing pertaining to the cartridge case under contract and all drawings and specifications listed thereon shall form a part of this specification.

A–2. All drawings and specifications shall be of the issue in effect on the date of invitation for bids.

B. TYPES

B–1. This specification covers the following types of one piece steel cartridge cases for artillery ammunition:

57-mm, M30
75-mm, M31

C. MATERIAL

C–1. *Material.*—The steel used for the manufacture of cartridge cases shall be in accordance with applicable specification or authorized by the Office, Chief of Ordnance.

D. GENERAL REQUIREMENTS

D–1. Cartridge cases shall meet all requirements prescribed on drawings and in applicable specifications.

D–2. Cartridge cases shall be manufactured in accordance with methods approved by the Office, Chief of Ordnance. Both the 57-mm, M30, cases and the 75-mm, M31, cases shall be heat treated.

D–3. Cartridge cases shall be sound and free from folds, cold shuts, scores, seams, and other injurious defects. Small dents, or minor defects, shall not be cause for rejections. In case of dispute the inspector shall be guided by samples of permissible defects furnished or approved by the contracting officer.

D–4. The flow of metal in forming the head shall be parallel to the curved inner surface joining the head and lower side wall so that no folds indicative of shear or any roughness exist.

D–5. The treatment and machining operations for each lot of cartridge cases shall be uniform.

D–6. Cold worked cases shall be stress relieved at a temperature of not less than 400°F. for 1 hour and the heat treated cases shall be drawn by subjecting the cases to a temperature of not less than 700°F. for 1 hour. Any other temperature-time cycle for conditioning the cases shall be approved by the Chief of Ordnance.

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E. DETAIL REQUIREMENTS

E–1. *Protective coatings.*

E–1*a*. Cartridge cases shall have a protective finish, either of treated zinc plate in accordance with type I, class RSC, of Specification No. 57–0–2, or of phenol formaldehyde varnish complying with Ordnance Department Tentative Specification AXS–736.

E–1*b*. Prior to the application of the varnish, the cases shall be given a surface conditioning treatment by a method conforming to grade II, type 4 or 5, Ordnance Department Tentative Specification AXS–1245, so that the cases will meet the applicable requirements of that specification at the time of varnishing.

E–1*c*. The varnish coating shall have a minimum thickness of .0003 inch in all portions of the cases excepting the sharp edges.

E–1*d*. If the finish is not treated zinc plate, the primer cavity shall be coated with either phenolic varnish or nitrocellulose lacquer. If phenolic varnish is used, the maximum thickness shall not exceed .0006 inch. If the phenolic varnish is not used, the primer cavity shall be coated after the baking of the varnish coating with nitrocellulose lacquer conforming to the requirements of class A, U. S. Army Specification 3–158, except that it may be colored with a soluble dye.

E–1*e*. When the unperforated section of the bodies of varnished cases are subjected to the abrasion resistance test (par. F–7*b*.), the area abraded through to the metal shall be less than 4-mm in diameter.

E–1*f*. When subjected to the salt spray test (par. E–7*a*), the varnished cartridge cases (without liners) shall show no signs of rust except in the sharp corners between flange and sidewall, on the sharp edges and the machined face of the base.

E–1*g*. Cases with zinc plate finish shall be subjected to a heat treatment of approximately 400°F. for 1 hour prior to the chromate treatment to avoid hydrogen embrittlement.

E–2. *Marking for identification.*—Each case shall be marked with the nomenclature of caliber, lot number, initials of manufacturer, and year of manufacture. The identification shall be either stamped on the base of the case with a metal stamp or stenciled on the base or side of the case, between flange and perforations, with an indelible ink. Stenciling shall be legible after salt spray test.

E–3. *Proving ground tests.*—The cartridge cases shall meet the following requirements in the proving ground tests:

E–3*a*. After firing, the cartridge cases shall be free of any cracks or ruptures.

E–3*b*. No primer shall be dislodged from the primer seat during firing or extraction of the case as evidenced by no protrusion of the primer head above the base of the case.

E–3*c*. No case fired in a gun shall be so deformed that it cannot readily be removed by finger pressure.

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F. METHOD OF SAMPLING, INSPECTION, AND TESTS

F–1. *Inspection.*

F–1*a*. Each cartridge case shall be inspected for defects such as seams, scores, folds, minute breaks in metal, roughness, or stress raisers, sharp edges, burrs around perforations on inside of case, and improper protective finish.

F–1*b*. *Gaging.*—In addition to the other dimensional gaging operations, the contractor shall gage each cartridge case in the maximum profile gage pertaining to the cartridge case under contract before the application of the protective coating. If the cartridge case fails to enter or withdraw from the maximum profile gage freely, it shall be rejected subject to correction and retest.

Note.—Drawings of the Government profile gage will be furnished to the contractor for his information upon application to the inspector.

F–1*c*. Sufficient number of finished cases of each lot shall be checked by the contractor in the maximum profile gage after the application of the protective coating to assure that the cases still will enter and withdraw freely.

F–1*d*. The thickness of varnish coating on the cases shall be determined when required by the inspector.

F–2. *Lot.*

F–2*a*. A lot of cases shall not exceed 20,000.

F–2*b*. The first lot manufactured on a contract shall not exceed 5,000 cases for all sizes except when the new contract immediately follows the manufacture of satisfactory cases of the same type, in which event the maximum size of the first lot shall be as prescribed in paragraph F–2*a*.

E–2*c*. A lot shall consist of cases fabricated by the same processes and generally from the same lot of cups or disks.

F–3. *Sampling.*—Forty-four sample cases shall be selected by the inspector from each lot.

F–4. *Number of tests.*—The sample cases selected shall be divided for tests as follows:

Hardness	20
Macro	20*
Salt spray	2
Abrasion	2
Proving ground	20

*Taken from samples tested for hardness.

F–5. *Hardness tests.*

F–5*a*. Tests shall be conducted in accordance with Federal Specification QQ–M–151 to determine compliance with the mandatory hardness values specified on the drawings and to check the values shown as advisory.

E–5*b*. Tests shall be made on a longitudinal section cut from the selected cases, the center line of the cartridge case to be the center line of the section. Readings shall be taken in the zones and on the surfaces designated on the drawings after being suitably prepared.

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F–5*c*. If the hardness of any case fails to meet the specified requirements, the lot shall be rejected.

F–6. *Macro examination.*—The curved portion of the cartridge case between the head and sidewall of the sections used in the hardness tests shall be carefully examined for folds and roughness. One of these sections shall be macro etched to show the grain flow pattern. If the samples contain any folds or roughness, or unsatisfactory flow pattern, the lot shall be rejected.

F–7. *Protective coating tests.*

F–7*a*. *Salt spray test.*—The two cartridge cases selected shall be subjected to the salt spray test outlined in Specification No. 57–0–2. For varnished cases, the test period shall be 24 continuous hours and for zinc plated cases, the period shall be as prescribed for the type I, class RSC, finish. At the completion of the test, the cases shall be examined for corrosion. Compliance with the applicable requirements of Specification No. 57–0–2 for zinc plated cases and compliance with paragraph E–1*f* and legibility of stenciling as required in paragraph E–2 for varnished cases shall be noted.

F–7*b*. *Abrasion resistance test.*

F–7*b*(1). The apparatus used in this test shall conform to the following:

The apparatus consists of a metal tube 3 feet long, having an inside diameter of ¾ inch. This tube is supported in a vertical position over a dish or other receptacle. The abrasive material, Ottawa sand (20 to 30 mesh), is placed in a funnel which fits snugly over the otuside of the tube. The flow is started and stopped by a thin metal disk which slides in and out of a slot cut in a tube 3 inches from the top. When the sand reaches the bottom of the tube, it scatters itself, to strifle a surface on the panel about 1 inch in diameter. The sand is constantly poured in until the required amount has been used.

F–7*b*(2). The specimens for the abrasion test shall be cylindrical sections cut from the finished cartridge cases at a 45° angle to the longitudinal axis of the case. The specimen is placed so that the area to be tested is directly below the opening of the tube and so that the distance from the tube and so that the distance from the tube to the specimen at the nearest point is 1 inch. The diameter of the area abraded is measured after 3 liters of clean fresh sand have been allowed to fall through the tube on to the test specimen. The tests shall be performed on both the outside and the inside surfaces of each specimen.

F–7*c*. *Retests.*—If any case fails to meet the requirements of the protective coating tests, a retest using three times the number of samples, may be made upon request of the contractor. If any case used in the retest fails to meet the requirements of paragraph E–1, the lot shall be rejected.

F–8. *Proving ground tests.*

F–8*a*. The sample cases selected under paragraph F–4 shall be assembled into complete rounds in the same manner as required

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for service firing. The cases shall be fired using a powder charge which will give the chamber pressure prescribed on the drawing of the cartridge case.

Note A. The powder used for these tests shall be the same as prescribed for the service round.

Note B. The projectile shall be asembled to the cartridge case using the same type crimp which is prescribed for the service round.

F–8*b*. *Retest.*

F–8*b*(1). A retest using double the number of sample cartridge cases specified in paragraph F–4 may be made if requested by the contractor, provided not more than one cartridge case fails to meet the proving ground requirements, paragraph E–3, except if in the opinion of the Proof Officer the cartridge case failure is due to a dimensional or visual defect, he shall recommend that the retest be allowed only after 100 percent inspection by the contractor for this defect.

F–8*b*(2). If any cartridge case fired in the retest fails to meet the requirements of paragraph E–3, the lot shall be rejected.

F–9. *Reworking.*—The contractor may request in writing to the contracting officer permission to rework once any rejected lot of cases, stating the procedure to be used to eliminate the rejectable defect observed on the original test. If permission is granted, the number of sample reworked cases submitted on the ballistic test shall be double the number of cases in the original test, unless otherwise directed by the Chief of Ordnance.

G. PACKING AND MARKING

G–1. *Packing.*—Unless otherwise specified, the cartridge cases shall be delivered in standard commercial containers so constructed as to insure acceptance by common or other carrier, for safe transportation, at the lowest rate, to the point of delivery. Lot number of contents shall be marked plainly on each container. The gross weight of any container and contents shall not exceed 70 pounds when packed in cardboard container or 100 pounds when packed in a wood or plywood container.

G–2. *Data cards.*—Data cards, 5 x 8 inches, made of commercial manila tag board, approximately 200 pounds ream weight, shall be prepared for each lot of accepted cartridge cases to provide the information specified below. Quantities (approximately 50 cards per lot) and distribution of data cards shall be in accordance with instructions issued by the Office, Chief of Ordnance. Under "Remarks" the name of the cup or disk supplier shall be indicated. Where reworked or reinspected lots of cases are submitted, the data cards shall show the procedure used in reworking, or the type of defect and number of defective cases rejected upon reinspection. Included should be the reference wherein permission for reworking or reinspection was granted.

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AMMUNITION DATA CARD

Kind .....Ammunition Lot No. .....

Quantity in Lot .....

Dwg. No.* .....Spec. No.* .....

Manufactured by .....Date .....

Contract No. .....Order No. .....

Packed .....in a box.

Remarks .....

Certified to by....., Inspector
(Print here name of inspector, district, arsenal, or loading plant, and subdistrict, where applicable)

*Include revision, amendments and ANR's.

G–2*a*. Information recorded under "Remarks" shall also include the following information:

a. Steel manufactured by .....

b. Cups manufactured by .....

c. *Method of manufacture*.—Identify the flow-sheet date, code letter, or number which gives complete details of the manufacturing process in producing the cartridge cases referred to on the Ammunition Data Card.

G–2*b*. Hardness values shall be recorded on the reverse side of each data card, listing the maximum, minimum, and average values attained at each specified location as prescribed on the drawing.

G–3. *Marking*.—The marking for shipment shall be in accordance with Specification No. 100–2.

H. NOTES

H–1. This specification replaces Ordnance Department Tentative Specification AXS–1515.

H–2. The use of this specification, whenever applicable, is mandatory on all procuring agencies of the Army.

Notice.—This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions, is intended for use only in connection with procurement by the United States Government; and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

[50–46–5]

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